The Overview of the Technology :

C-MET has developed low dielectric and low loss ceramic filled polymer substrates suitable for microwave circuit design through a patented process. Copper cladding technology has been established to realize fine circuitry over the flexible substrates. These niche products exhibit excellent dimensional stability and tight dielectric tolerance suitable for high end microwave circuit design.

Salient features of the indigenously developed substrates compared to commercially available ones are:

- Ultra low loss tangent for selectivity
- Low dielectric anisotropy and tight dielectric tolerance
- Excellent temperature stability for outdoor wireless applications

Typical properties of the microwave substrates include:

Dielectric constant : 6.15±0.15 and 3.0±0.03 at 10 GHz

Loss tangent : 0.0018 to 0.002 at X-band

Temperature coefficient of dielectric constant : -18 to +25 ppm/^{o}C

Ultimate tensile strength: >14 MPa

Moisture absorption : 0.05-0.08%



Low dielectric microwave substrates developed at C-MET

The level of technology readiness at C-MET

The technology readiness level is TRL6

Period of training under the agreement

C-MET will provide 30 days in-house training on the fabrication of 10 Nos. of 8”x8” size microwave substrate will be demonstrated.

The ToT package contains the following

1. Document containing raw material details
2. Processing details
3. Product specification
4. Test plan and procedure
5. Training

PIEZOCERAMIC COMPOSITIONS and COMPONENTS

The Overview of the Technology :

Centre for Materials for Electronics Technology (C-MET) , Thrissur has developed the following piezoceramic compositions which are engineered for enhanced electromechanical properties. Different standard compositions, as shown below are available. C-MET also has the capability for custom-made piezoceramic compositions to suit the requirement of a particular application.

Piezoceramic compositions are typically suited for applications which include;

- Piezoactuators
- SONARS
- Hydrophones
- Other Ultrasonic applications

Standard Piezoceramic compositions

Sl.No	Product	Code
1	Piezoceramic composition	PZT-H1
2	Piezoceramic composition	PZT-H2
3	Piezoceramic composition	PZT-S1
4	Piezoceramic composition	PZT-S2

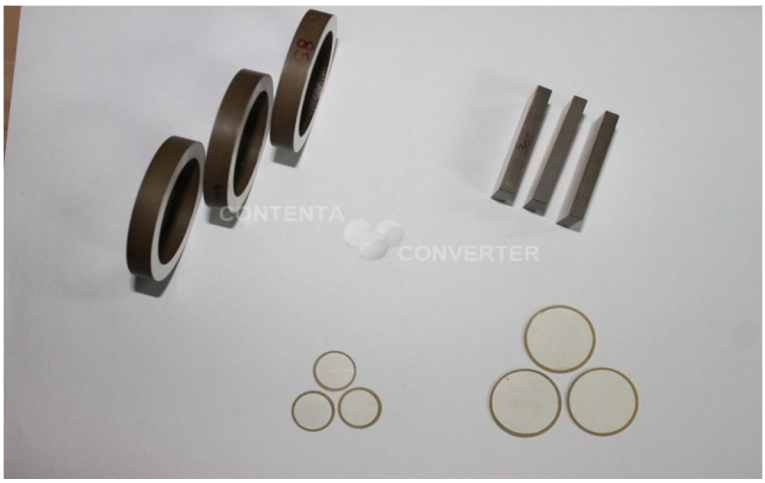
The important properties of standard piezoceramics offered by C-MET are given in **Table 1** given below.

Table 1 Specifications of Standard Piezoceramic Compositions

Electrical Characteristics		PZT-H1	PZT-H2	PZT-S1	PZT-S2
1	Dielectric Permittivity,ε_r (±5%)	1430	1120	1743	2682
2	Dielectric loss ,tan δ (±0.001)	0.003	0.002	0.02	0.02
3	Piezoelectric Coefft.d_{33}(pC/N) (±5%)	355	328	456	630
4	Electromechanical coupling factor (k_p) (±3%)	0.59	0.58	0.64	0.65
5	Mechanical quality factor Q_m (±10%)	685	900	72	70
6	Density (gm./cc)	7.6	7.6	7.7	7.6
7	Frequency constant,N Hz.-m (±1%)	2083	2100	1900	1975

Piezoceramic Components

C-MET has the capability to fabricate Components of various shapes. Typical dimensions and some of the important properties of piezoceramic components offered by C-MET are given in Table 2.



Piezoceramic Components – PZT Rings, PZT Plate and PZT Discs

Advantages

- Lower sintering temperature
- High-density microstructures
- Enhanced electromechanical properties
- Improved Insulation Resistance(IR) in Components

Table 2. Specifications of Piezoceramic Components

Piezoceramic component fabricated using specific piezoceramic composition	**Typical dimensions**	**Electrical Characteristics**					
		C(nF)	**tan δ**	**d_{33} (pC/N)**	**f_r (KHz)**	**k_{eff}**	**IR (GΩ)**
PZT Ring	OD = 50.00±0.05mm ID = 38.00±0.10mm T = 8.00 ±0.08mm	1.15± 0.04	0.002	325±25	179±1	0.58± 0.02	150±25 (1000V DC)
PZT Disc	Dia = 15.04±0.06mm T = 0.28 mm	10.50± 0.50	0.02	520±20	135±1	0.65± 0.02	20 (100V DC)
PZT Disc	Dia = 24.70±0.20mm T = 1.03±0.03mm	10.0± 1.0	0.02	575±30	81.75± 0.75	0.56± 0.02	20 (100V DC)
PZT Plate	Length = 40.00± 0.03 Width = 19.94±0.02 T = 5.00 ±0.01	1.60± 0.07	0.002	275±50	394±2	0.58± 0.02	110±10 (1000 V DC)

The level of technology readiness at C-MET

The technology readiness level is 6

Period of training under the agreement

C-MET will provide two weeks training on the development of piezoelectric composition (1kg batch size) at the premises of C-MET and making of 10 numbers of corresponding piezoceramic component will be demonstrated.

The ToT package contains the following

1. Document containing development of piezoelectric composition
2. Fabrication details of piezoceramic component
3. Product specification
4. Test plan and procedure
5. Training